

AI Worksheet: Classification Metrics and Error Analysis

Context: In the Educational Companion System, supervised models and rules may classify learner needs (e.g., remediation vs enrichment). This worksheet practices model evaluation using a confusion matrix, precision, recall, F1-score, and threshold trade-offs. Original demo content for thesis ingestion (selectable text).

Intended learning outcomes

1) Build and interpret a binary confusion matrix. 2) Compute accuracy, precision, recall, and F1. 3) Explain when accuracy is misleading. 4) Relate threshold changes to false positives and false negatives. 5) Plan a short error analysis for one misclassified case.

Confusion matrix (binary classification)

Labels: Positive = learner needs intervention. Negative = on track.

TP = predicted positive, actually positive.

FP = predicted positive, actually negative (false alarm).

FN = predicted negative, actually positive (missed case).

TN = predicted negative, actually negative.

Worked example

Counts: TP=42, FN=8, FP=5, TN=145 (total N=200).

Accuracy = $(TP+TN)/N = 187/200 = 0.935$.

Precision = $TP/(TP+FP) = 42/47 = 0.894$ (when we alert, we are usually right).

Recall = $TP/(TP+FN) = 42/50 = 0.840$ (we catch 84% of true intervention cases).

F1 = $2 \cdot P \cdot R / (P + R) = 2 \cdot 0.894 \cdot 0.840 / (0.894 + 0.840) = 0.866$.

Interpretation

High precision protects learners from unnecessary nudges. High recall protects at-risk learners from being overlooked. For educational recommendations, false negatives may be costlier than false positives depending on policy.

Threshold trade-off

Lowering the decision threshold usually increases recall and FP. Raising it increases precision and FN. Plot precision-recall for your dev set before choosing an operating point. Document the chosen threshold in the model card.

Error analysis

Pick one FP: why did the model think intervention was needed? Missing feature, stale activity log, or ambiguous label? Pick one FN: was activity not ingested yet? Propose one feature or rule change and how you would validate it.

Practice tasks

- 1) If FN cost is 3x FP cost, would you lower or raise threshold? Justify.
- 2) Recalculate metrics if TP=48 and FN=2 with same FP, TN.
- 3) When is accuracy misleading? Give class imbalance example.
- 4) List two keywords you expect in an extracted summary: confusion matrix, F1-score.

Reflection questions

How should access-aware filtering interact with evaluation (only labeled accessible items)? Why must ingestion summaries stay scoped to the parent resource visibility? How could richer PDF text improve semantic matching for AI units without exposing full files?

Keywords

classification metrics, confusion matrix, precision, recall, F1-score, error analysis, threshold, supervised learning, model evaluation.